$$\mathbf{L}|\lambda_1\rangle = \lambda_1|\lambda_1\rangle$$

$$\mathbf{L}|\lambda_2\rangle = \lambda_2|\lambda_2\rangle.$$

Now, using the fact that **L** is Hermitian (its own Hermitian conjugate), we can flip the first equation into a bra equation. Thus,

$$\langle\lambda_1|\mathbf{L} = \lambda_1\langle\lambda_1|$$

$$\mathbf{L}|\lambda_2\rangle = \lambda_2|\lambda_2\rangle.$$

By now, the trick should be obvious, but I will spell it out. Take the first equation and form its inner product with $|\lambda_2>$. Then, take the second equation and form its inner product with $<\lambda_1|$. The result is

$$\langle\lambda_1|\mathbf{L}|\lambda_2\rangle = \lambda_1\langle\lambda_1|\lambda_2\rangle$$

$$\langle\lambda_1|\mathbf{L}|\lambda_2\rangle = \lambda_2\langle\lambda_1|\lambda_2\rangle.$$

By subtracting, we get

$$(\lambda_1 - \lambda_2)\langle\lambda_1|\lambda_2\rangle = 0.$$

Therefore, if λ_1 and λ_2 are different, the inner product $<\lambda_1|\lambda_2>$ must be zero. In other words, the two eigenvectors must be orthogonal.

Next, let's prove that even if $\lambda_1 = \lambda_2$, the two eigenvectors can be *chosen* to be orthogonal. Suppose

$$\mathbf{L}|\lambda_1\rangle = \lambda|\lambda_1\rangle$$

$$\mathbf{L}|\lambda_2\rangle = \lambda|\lambda_2\rangle. \qquad (3.10)$$